

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

■ Operating Systems ■ CS-3009 ■ Autumn-2022, Mid Sem

■ 30 Marks ■ Time: 2hrs ■ 1 Page ■ Answer ALL Questions

1. (a) Explain convoy effect with suitable example. [2]
- (b) What do you mean by SPOOLing? With a neat diagram, explain how SPOOLing works. [2]
- (c) Compare the message passing IPC with shared memory IPC with their advantages and disadvantages. [2]
- (d) Suppose, being an OS developer, your objective is to minimize the total time to complete a given set of processes. Consider that there are two processes: A and B. Can there be a scenario where purely sequential execution of the processes will perform better compared to executing them in an interleaved manner with context switching? Justify your answer. [2]
- (e) What is *Race Condition*? The following two processes P1 and P2 that share a variable B with an initial value of 2 execute concurrently. What are the distinct values B can possibly have after the execution? [2]

Process P1

```
{  
    C = B - 1;  
    B = 2 × C;  
}
```

Process P2

```
{  
    D = 2 × B;  
    B = D - 1;  
}
```

2. (a) Differentiate between preemptive and non-preemptive scheduling. An operating system uses the Longest Remaining Time First (LRTF) process scheduling algorithm, in contrast to Shortest Remaining Time First (SRTF) algorithm. Consider the set of Processes [A, B, C, D]; Arrival Times [1, 2, 3, 4]; Burst Times [5, 1, 3, 1]. For tie breaking OS first focuses on minimizing context switch overhead then on the process close to the front end of the ready queue. Draw the Gantt chart. Find the average waiting time of all the processes. Also find the CPU utilization. [4]
- (b) Consider the following solution to critical section problem, for two processes. Explain if this solution satisfies mutual exclusion, bounded waiting, and progress property or not. [6]

Code for $P_i; i \in \{1, 2\}$

```
repeat{  
    inside[i] = true;  
    while(inside[j]){  
        inside[i] = false;  
        while(turn == j) continue;  
        inside[i] = true;  
    }  
    ...Critical Section ...  
    turn = j;  
    inside[i] = false;  
    ...Reminder Section ...  
} until false
```

3. (a) Consider a preemptive CPU scheduling algorithm named *Minimum CPU Used First* (MCUF). This algorithm compares the processor time used/executed by the processes so far and selects the process that has used the CPU time for minimum. If there is a tie then it can be resolved with the following tie breaking rules, in sequel: (i) avoid context switch (ii) minimum remaining time, (iii) close to the front end of the ready queue.
- Consider 5 processes; [A, B, C, D, E] with arrival times [0, 1, 4, 6, 8] and burst times [4, 5, 3, 2, 4]. Find the schedule using MCUF algorithm. Show the Gantt chart and the intermediate steps clearly. The scheduling points are integer points only. Find the waiting time of each process. Briefly explain if there can be starvation to some processes. [7]
- (b) Consider a system with 'm' CPUs and 'n' processes where $m < n$, then what will be the upper bound and lower bound on number of processes that may be present in the running state, ready state, and waiting state respectively? Justify. [3]